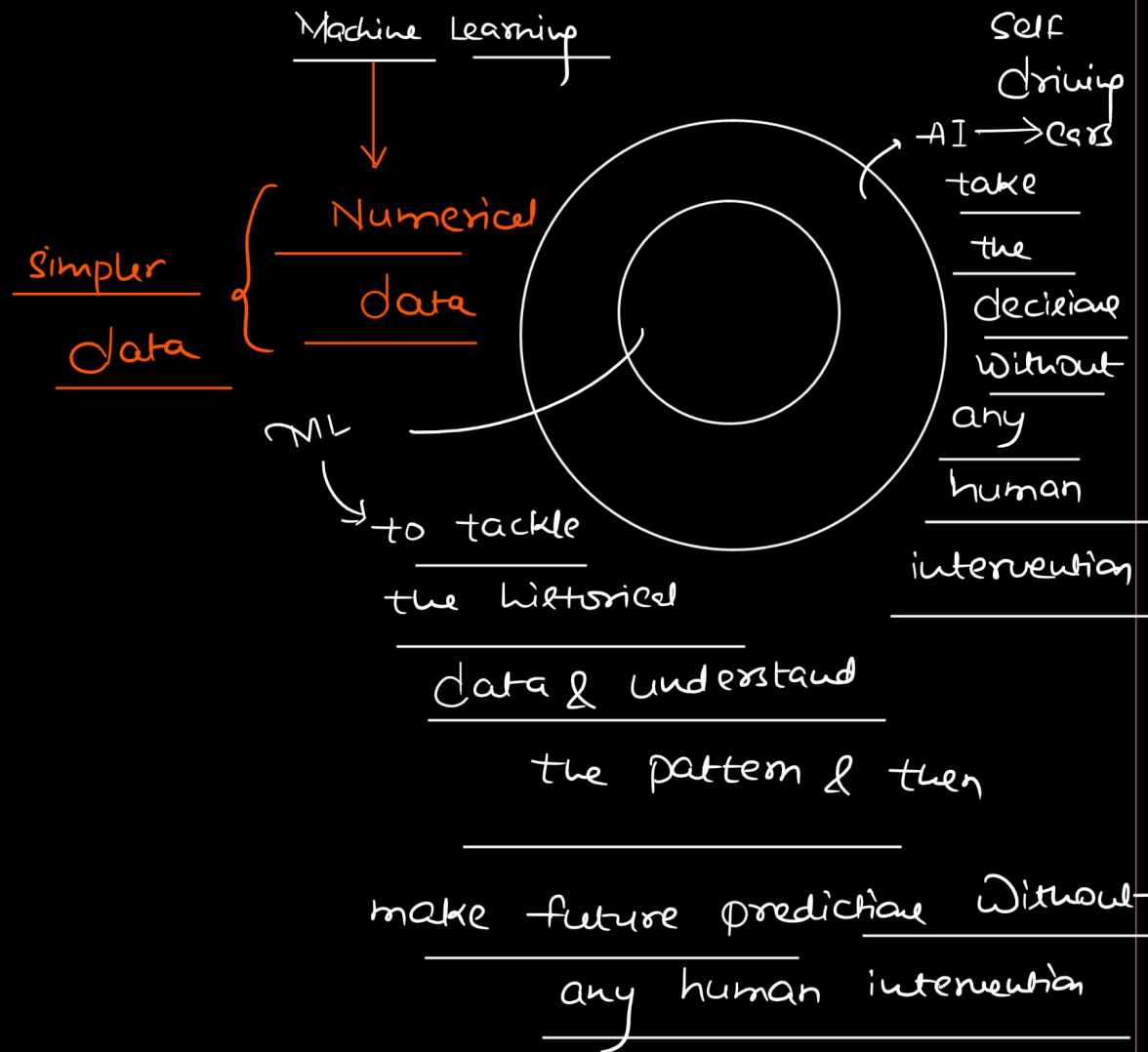
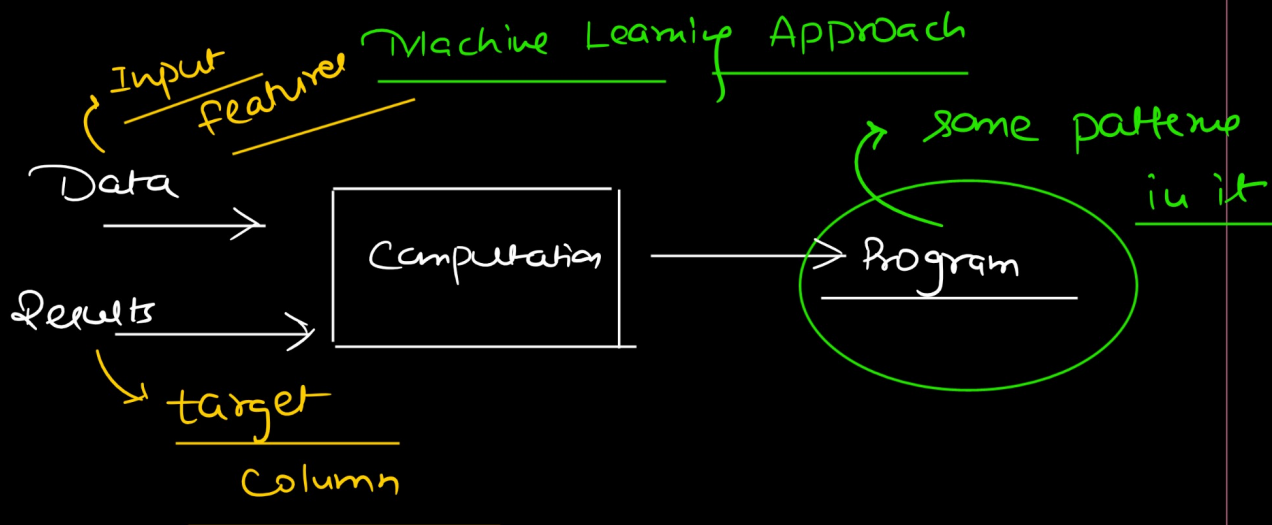
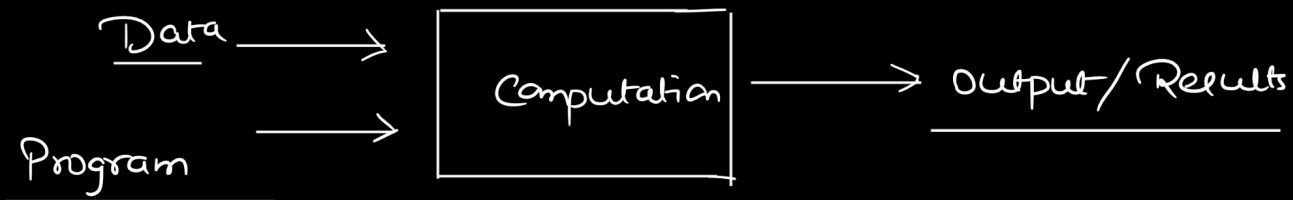
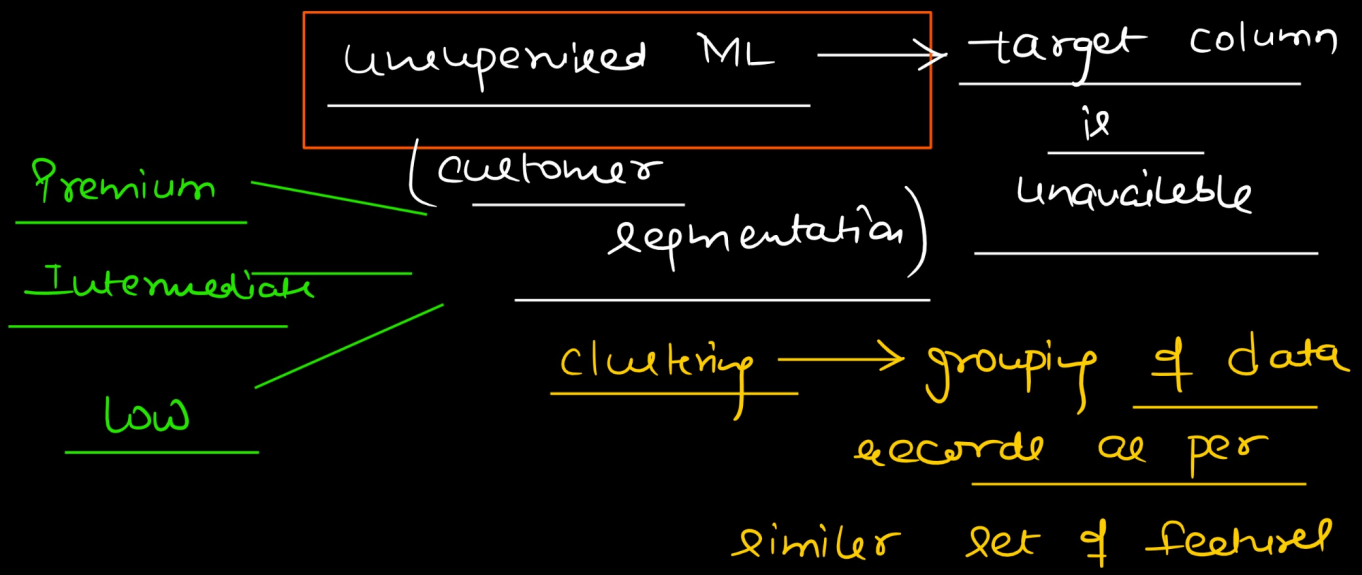
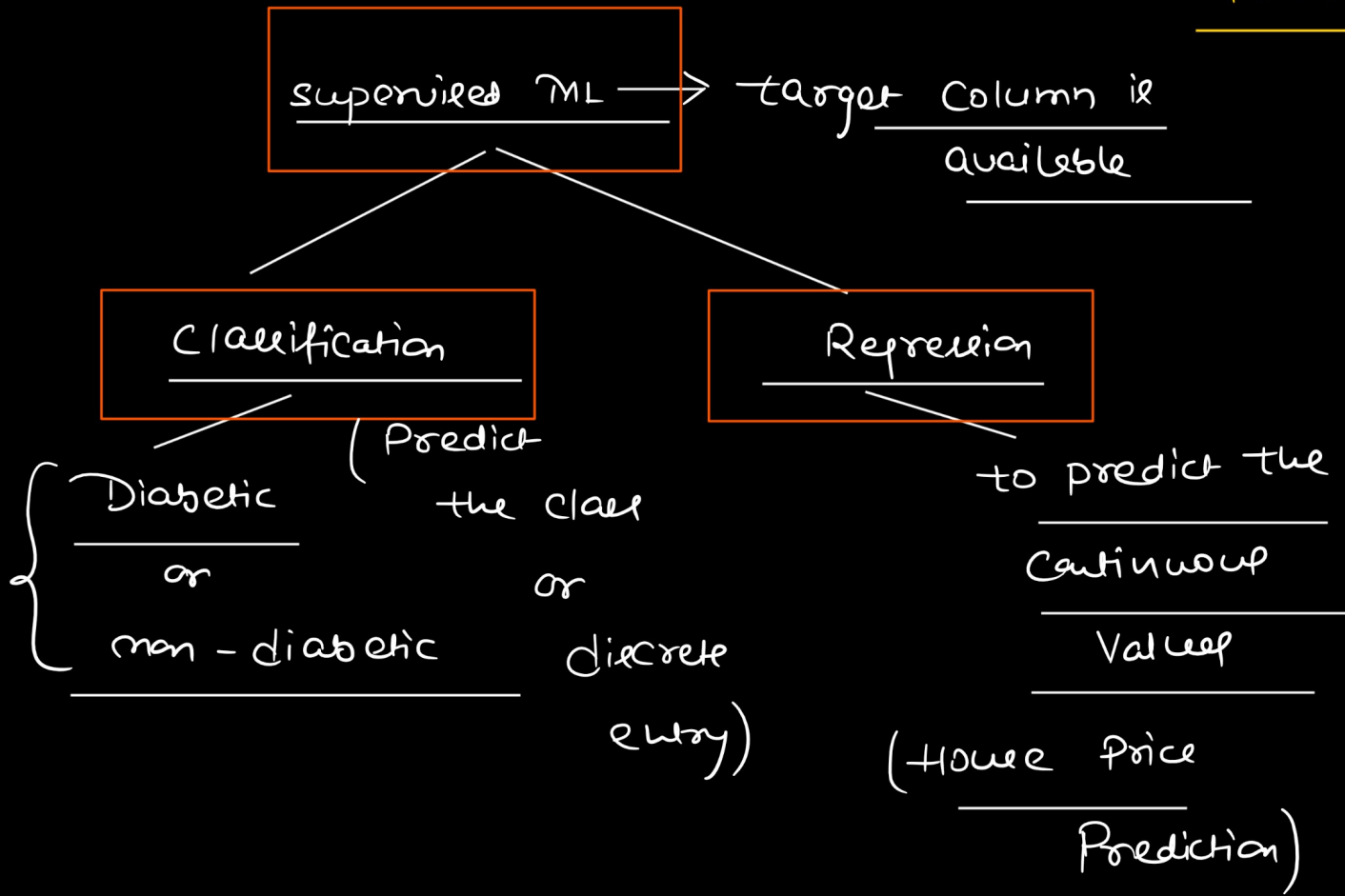




Traditional Programming





① Problem Definition



{ Statistics ← (3) Data Preprocess → clean,
Python Programming transforming,
 { ↳ Numpy splitting the
 ↳ Pandas data

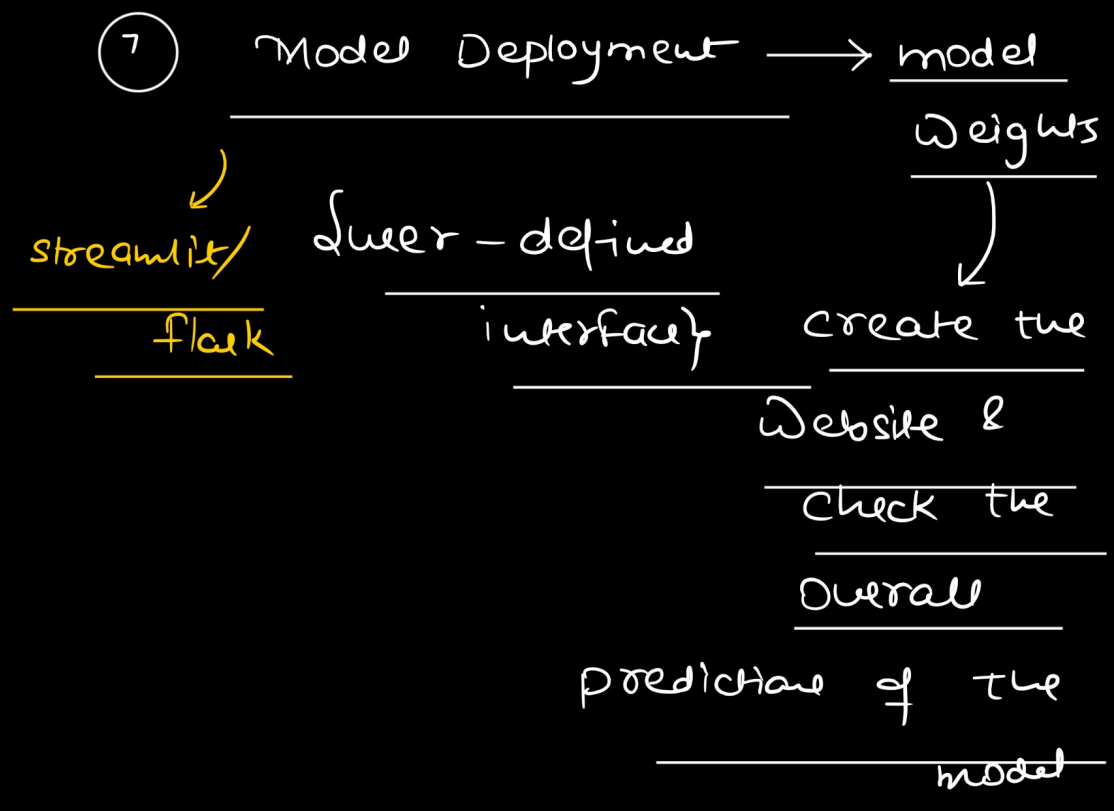
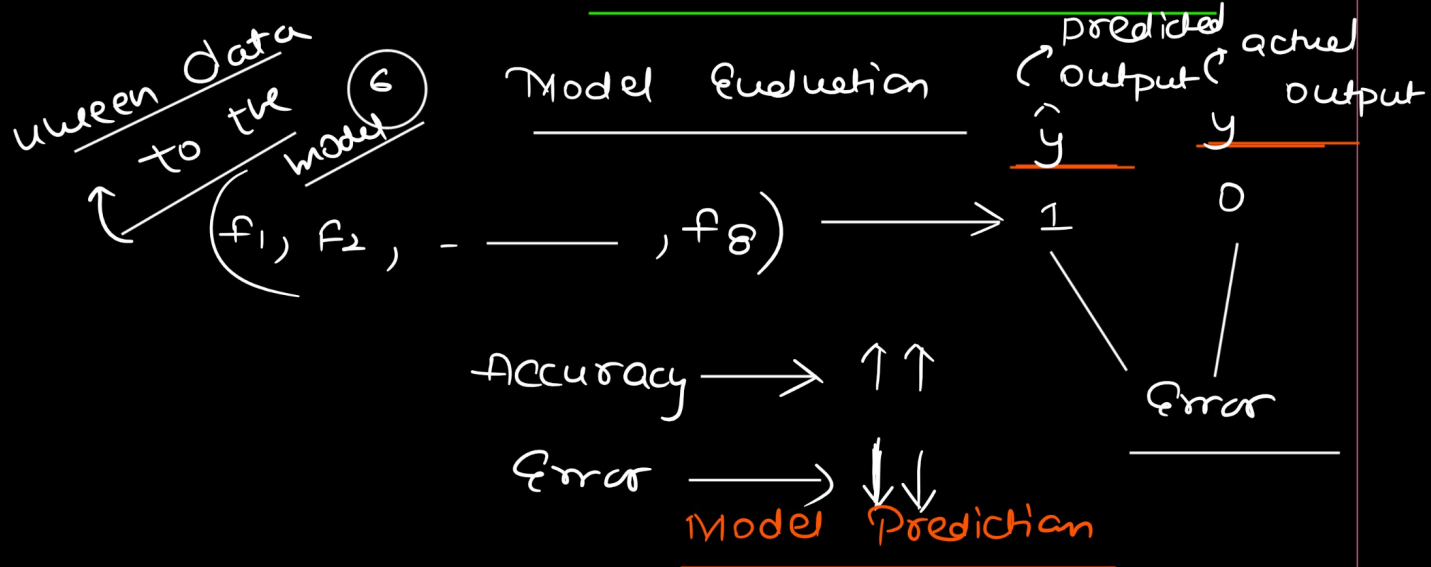
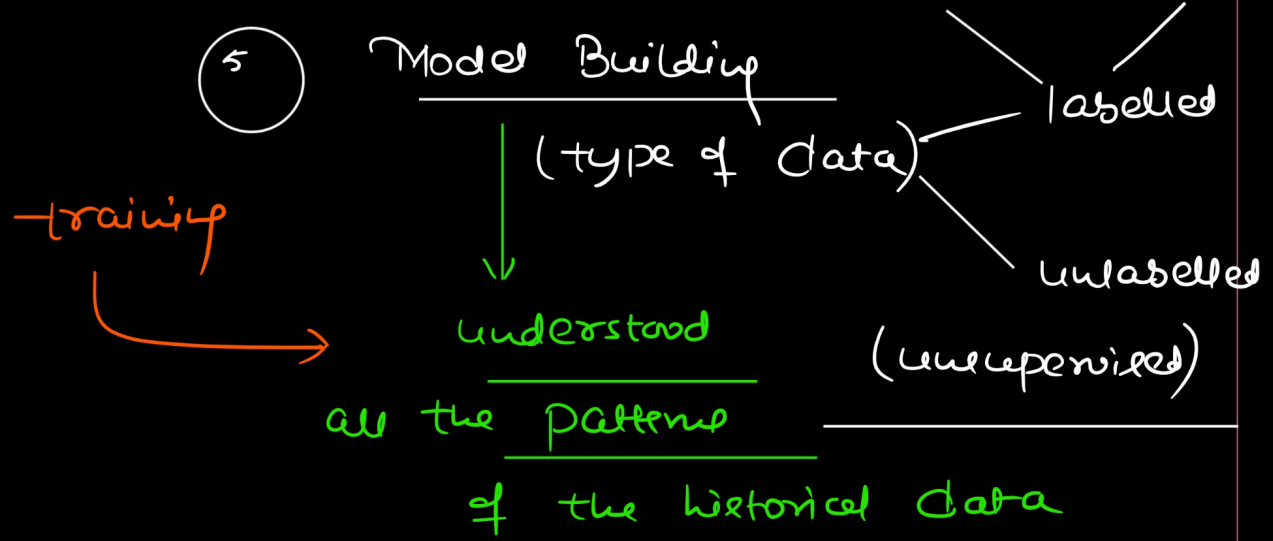
Understand the patterns
& relationships

(supervised)

Continuous categorical

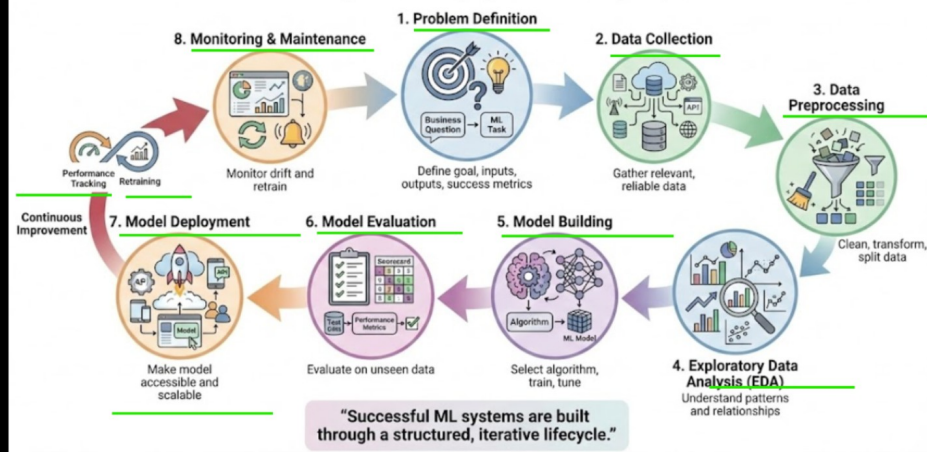
4 sklearn $\leftarrow$ ML

(Preprocessing,
Algorithm,
train test split)



The Machine Learning Project Lifecycle

From business problem to deployed and evolving AI system



xx

Pretrained Models ← Generative AI

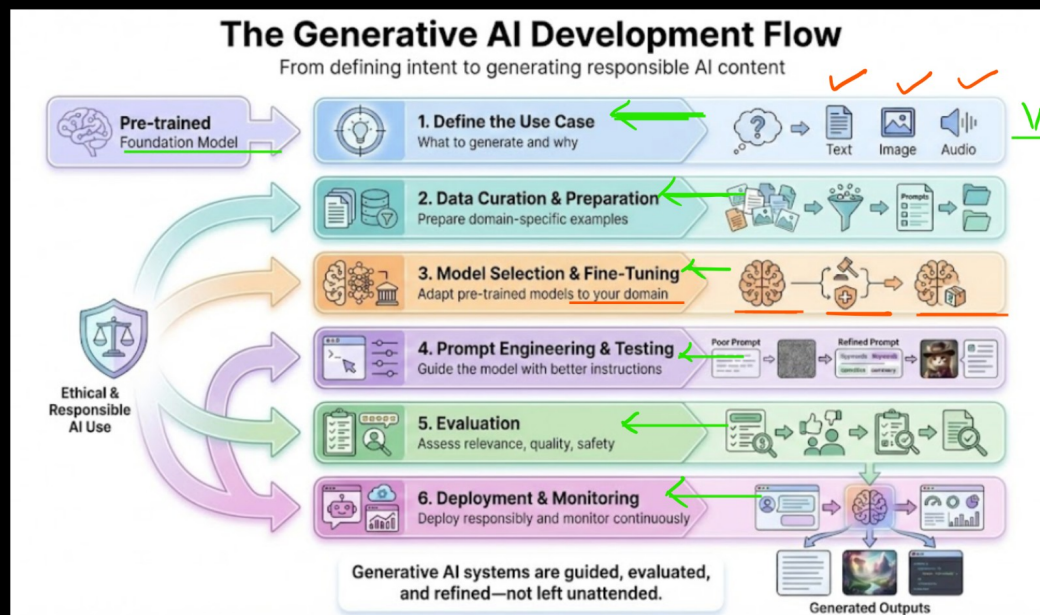
already trained on a very huge amount of data

generating the new content
text, image, music
based on patterns it has learned

fine tune → specific purpose

Pretrained models are artificial intelligence models previously trained on massive, general-purpose datasets (e.g., ImageNet, Wikipedia) to learn foundational features, allowing them to be reused or fine-tuned for specific, related tasks.

Fine-tuning in generative AI is the process of further training a pre-trained, general-purpose foundation model (like GPT or Llama) on a smaller, domain-specific dataset to enhance its accuracy, tone, and specialization for niche tasks.



Doctor GPT

"DoctorGPT" refers to several distinct AI-driven medical and technical projects, ranging from clinical assistants for healthcare professionals to software debugging tools.

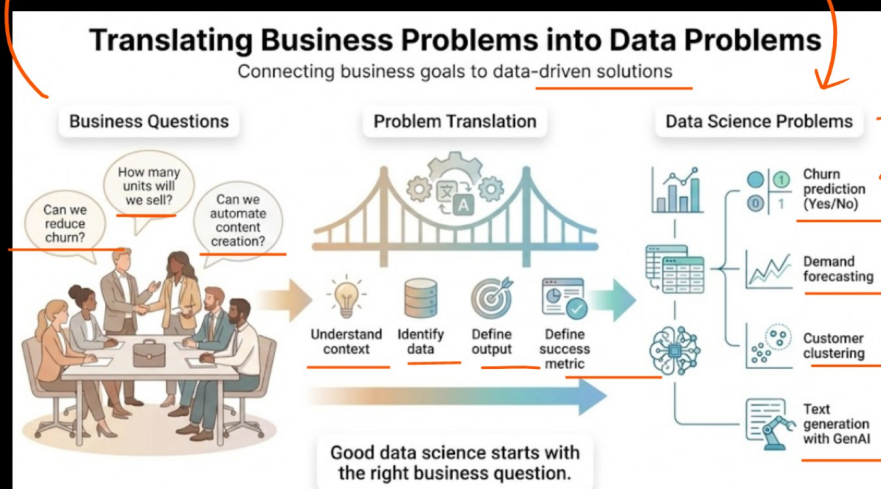
Translate business problem into

data problem

↳ measurable

solvable

using data



Domain expertise

Steps to translate effectively:

- 
1. Understand the business context — What decision will this help improve?
 2. Identify data sources — What data do we already have, and what do we need?
 3. Clarify the output — Are we predicting, classifying, or generating something?
 4. Define success — What metric will show the model is useful (accuracy, revenue impact, etc.)?